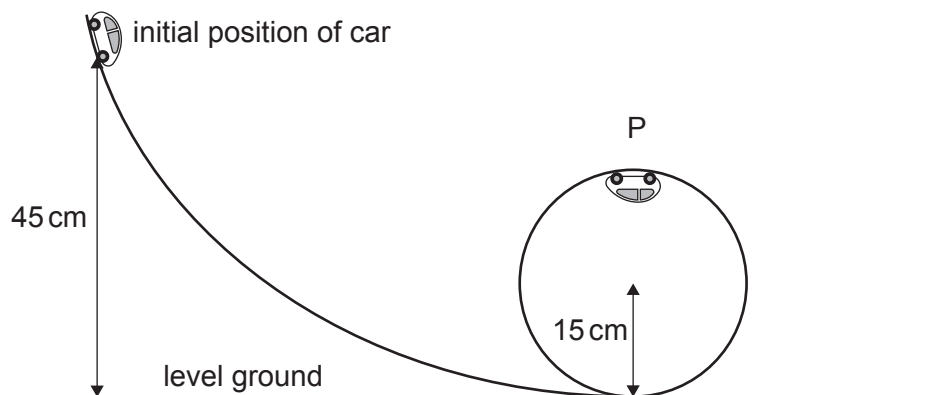


12. (a) In a “loop-the-loop” toy, a car of mass 0.12 kg is released from rest. The initial position of the car is 45 cm above level ground. The radius of the circular loop is 15 cm . The car reaches the top of the loop at position P. Frictional and air resistance forces are negligible.

diagram not to scale



- (i) Show that the speed of the car at P is 1.7 ms^{-1} . [2]

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- (ii) Determine the normal force exerted by the loop on the car at P. [3]

.....

- (iii) State why the car stays in contact with the loop. [1]

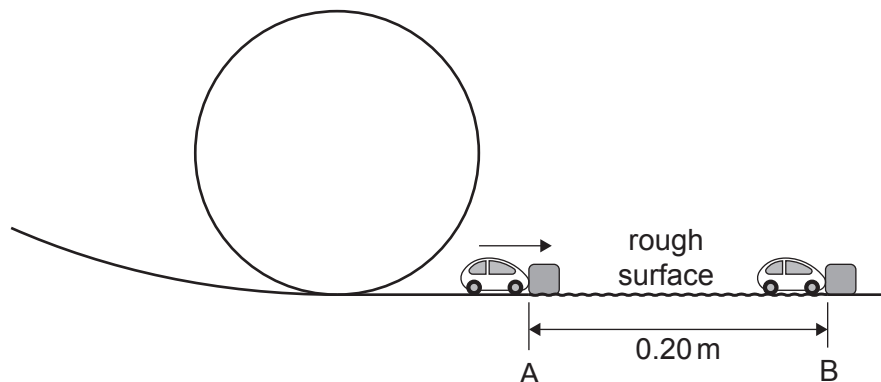
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(Question 12 continued)

- (b) At point A the car collides with a block of mass 0.18 kg and sticks to it. After the collision, the car and the block move together with speed 1.2 ms^{-1} .

diagram not to scale



- (i) Calculate the speed of the car just before it collides with the block.

[2]

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- (ii) The surface from A to B is rough and the combined car and block come to rest at B. The distance AB is 0.20 m . Determine the rate of change of momentum of the combined car and block from A to B.

[3]

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(Question 12 continued)

- (iii) Calculate the dynamic coefficient of friction between the rough surface and the combined car and block.

[2]